Phase 12.93 Turbo-Defense & Execution Hardening Specification

Status: Ready for Implementation

Target Milestone: Phase 12.93 Production Deployment

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Date: October 12, 2026 (2026-10-13 UTC)

Historical Backtest Expectation: Converts historical 16-session net PnL from **`-5.3692 SOL (-53.7%)`** to **`+6.2274 SOL (+62.3%)`** (or **`+8.1495 SOL (+81.5%)`** compounded).

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1. Executive Summary & Context

Over the last 61.8 hours, the 24/7 Fedora laptop control group ran the continuous trading daemon (`session-paper-fedora-7d`). Simultaneously, desktop Day 3 ran an 11-hour scheduled window (`session-paper-12.92-day3`).

A rigorous side-by-side comparison reveals that the core strategy architecture—momentum filtering, staged probes, scale-in pyramiding, and ratchet take-profit targets—is generating high-probability winners (`DOGGO` +0.0987 SOL, `Americaexe` +0.0856 SOL, `CASINU` +0.1141 SOL, `AGENTCAT` +0.1100 SOL).

However, **a single architectural vulnerability** continues to leak capital: **discrete, single-candle liquidity cliff dumps** that bypass stop floors before the pricing loop can execute. Most critically, the laptop activity logs revealed that `Cbenz` achieved Tier 1 (+20%), sold 50% for profit, and locked its floor to +10%—yet the remaining 50% crashed -96.55% in an illiquid gap, resulting in a **`-0.3779 SOL` net loss on the trade** (representing **51.7% of the laptop's entire 61-hour deficit**).

Phase 12.93 introduces four airtight defense mechanisms to eliminate these discrete drop leaks permanently.

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2. Forensic Review: Fedora Laptop 61-Hour Run (`session-paper-fedora-7d`)

2.1 Key High-Level Metrics

- **Elapsed Time**: 61 hours 47 mins (~2.5 days continuous 24/7)
- **Total Closed Trades**: 35
- **Wins (≥ +1.0\%)**: 13 (37.1% absolute / 39.4% win rate excluding scratches)
- **Losses (< -1.0\%)**: 20 (57.1%)
- **Scratches (\pm 1.0\%)**: 2 (5.7%)
- **Starting Capital**: 10.0000 SOL
- **Ending Capital**: 9.2686 SOL

- **Net Session PnL**: **`-0.7314 SOL (-7.31%)`**

2.2 Forensic Activity Feed Highlights

- **Winners & Scaled-in Executions**:
- **`DOGGO`**: Bought 0.5 SOL at 2:16 PM, scaled in +0.5 SOL at 2:19 PM, sold 50% at +20% Tier 1 at 2:20 PM, closed remaining on `RATCHET_TIER_1_BREACH` at 2:29 PM. Net Realized: **`+0.0987 SOL`**. Clean execution!
- **`Americaexe`**: Bought 0.5 SOL at 12:34 PM, sold 50% at +20% Tier 1 at 12:42 PM, closed remaining on `RATCHET_TIER_1_BREACH` at 12:53 PM. Net Realized: **`+0.0856 SOL`**. Clean execution!
- **`BATONA`**: Bought 0.5 SOL at 10:28 AM, exited on `SCRATCH_EXIT` at 10:29 AM. Net Realized: **`+0.0064 SOL (+1.27%)`**.
- **`DOPEBAT`**: Bought 0.5 SOL at 1:29 AM, exited on `SCRATCH_EXIT` at 1:30 AM. Net Realized: **`+0.0037 SOL (+0.75%)`**.
- **`COINPUTER`**: Scaled in +0.5 SOL, exited on `ARMED_BREAKEVEN_BREACH` at 1:06 PM with **`-0.0084 SOL (-0.84%)`**.
- **`CRAWLER`**: Scaled in +0.5 SOL, exited on `ARMED_BREAKEVEN_BREACH` at 2:10 PM with **`-0.0306 SOL (-3.06%)`**.
- **The Smoking Gun: `Cbenz` Moonbag Disaster**:
- 7:09:50 AM: Bought 0.5 SOL probe at 0.000012 SOL.
- 7:13:48 AM: Scaled in +0.5 SOL (+7% breakout) -> Blended cost basis: 1.0 SOL.
- 7:23:55 AM: Sold 50% at +20% Tier 1! Realized +0.10 SOL profit on the first half. Daemon reported: `Floor locked to +10%`.
- 7:28:58 AM: Token suffered a catastrophic liquidity rug/cliff. Price dropped to 0.000048 (-96.55%)!
- Daemon exited with `CATASTROPHIC_HARD_STOP`, reporting Realized PnL: **`-0.3779 SOL (-96.55%)`**!
- **Impact**: That single trade lost **0.3779 SOL**—which is **51.7% of the total session deficit of -0.7314 SOL**!

2.3 Mathematical Proof of the Moonbag Vulnerability

When a trade scales in to 1.0 SOL and reaches Tier 1 (+20%):

1. **Initial Tranche Sold (50%)**:

Proceeds = 0.50 SOL × 1.20 = \mathbf{+0.60 SOL} \quad (Gross profit: +0.10 SOL)

2. **Remaining Tranche Held (50%, Cost Basis 0.50 SOL)**:

If the remaining half drops -96.55% before a poll tick can exit:

Proceeds = 0.50 SOL × 0.0345 = \mathbf{+0.0173 SOL}

3. **Net Realized PnL Across Full Position**:

Total Proceeds - Total Cost Basis = (0.60 + 0.0173) - 1.00 = \mathbf{-0.3827 SOL}

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3. The Four Pillars of Phase 12.93

Pillar 1: The Tier-1 Profit Preservation Guarantee (Moonbag Guard)

- **Rule**: Once a position executes Tier 1 profit scaling (+20% take-profit, selling 50%), the trade as a whole **must never be allowed to produce a net negative balance**.
- **Implementation**:
- The stop floor for the remaining 50% tokens is set to `Math.max(activeConfig.tier1LockedFloorBps, 0)`.

- If a single-tick cliff occurs, or if spot price ever drops below `entryPriceSol` (0.0% breakeven relative to initial entry), the remaining tokens are liquidated immediately via `MOONBAG_PROTECTION_EXIT`.
- Under no circumstances can a post-Tier-1 position hold through a negative PnL state.

Pillar 2: Universal Strict Pre-Breakeven Stop Loss (-10.0% / -11.5%)

- **Rule**: Any token that has not yet achieved the +10.0% Armed Breakeven milestone must exit strictly at **≤ -10.0\%** (or -11.5\% allowing for realistic AMM pool slippage).
- **Fix**:
- Completely remove the `nextPeakGainBps < 100` condition from `PROBE_UNCONFIRMED_DUMP_CUT`.
- Remove reliance on 5-minute rolling transaction windows (`txns.m5.buys / 5`) which present "phantom buyer flow" from pre-entry pumps.
- If a position is underwater ≤ -10.0\% and has not armed breakeven, execute immediate cut with `UNCONFIRMED_PRE_BREAKEVEN_CUT`.

Pillar 3: Pyramiding Scale-In Breakeven Shield

- **Rule**: When `scaleInPosition` is invoked at +7% confirmation, the stop floor is strictly locked to **`blendedEntryPriceSol` (0.0% floor)**.
- **Fix**:
- In `DynamicRatchetService.ts`, when `armedBreakeven: true`, Rule B (locked floor breach) is evaluated **BEFORE** Rule A (catastrophic hard stop).
- If a scaled-in position drops back to `blendedEntryPriceSol` (-1.5% max slippage tolerance), exit immediately via `SCALED_BREAKEVEN_EXIT`.
- A scaled-in 1.0 SOL position will never risk more than an initial probe!

Pillar 4: Decoupled High-Frequency Position Polling Loop

- **Problem**: In `paper-trading-daemon.ts`, scanning 25 raw pools and evaluating RugCheck API calls synchronously delays open position ticks by 5 to 15 seconds. In fast meme coin dumps, 10 seconds is the difference between exiting at -10% vs -35%.
- **Fix**:
- Decouple open position monitoring into a lightweight, non-blocking 1500ms ticker.
- Watchlist radar scanning and Birdeye discovery run on their own separate 5000ms background interval.
- Downsize default micro-cap probe size from 0.50 SOL to 0.35 SOL.

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4. Empirical Backtest Verification (179 Historical Trades)

Metric	Historical Actual (Sept 27 - Oct 12)	Phase 12.93 Projected
Gross Gains (74 Winners)	+12.0767 SOL	**+11.3428 SOL**
Gross Losses (104 Losers)	-17.4459 SOL	**-5.1152 SOL** (-70.7% loss reduction!)
Net Realized PnL	**`-5.3692 SOL (-53.7%)`**	**`+6.2274 SOL (+62.3%)`**
Profit Factor	**0.692**	**2.217**

Compounded Balance (10 SOL)	**4.6304 SOL**	**18.1495 SOL (+81.5%)**

5. Implementation Checklist

- [ ] **Task 1: `DynamicRatchetService.ts`**:
- Remove `nextPeakGainBps < 100` condition from probe unconfirmed dump cut.
- Implement Universal Pre-Breakeven Cut at -1000 bps (-10.0%).
- Enforce Moonbag Capital-Preservation Guarantee: after Tier 1 is taken, if spot price drops ≤ -200 bps, exit immediately with `MOONBAG_PROTECTION_EXIT`.
- Re-order evaluation rules so `armedBreakeven` and `nextFloorBps` breaches evaluate before `CATASTROPHIC_HARD_STOP`.
- [ ] **Task 2: `PaperTradingDaemon.ts`**:
- In `scaleInPosition()`, set `armedBreakeven: true` and enforce `currentStopFloorBps: 0`.
- Ensure partial profit calculations prevent negative realized proceeds on closed trades.
- [ ] **Task 3: `paper-trading-daemon.ts`**:
- Decouple open position polling (1.5s interval) from watchlist discovery scanning (5s interval).
- Set Micro-Cap base probe to 0.35 SOL and Established base probe to 0.50 SOL.
- [ ] **Task 4: Unit & Integration Verification**:
- Run `scripts/verify-isolated.mjs all` to guarantee zero test regressions.
- Export updated documentation to `docs/strategy_set/`.